Rayen Benabdelhafidh

TELECOMMUNICATIONS ENGINEERING & AI CLOUD ARCHITECT | CYBERSECURITY



SUMMARY

Telecommunications engineer with triple expertise in **Cloud/DevOps** (Kubernetes, AWS, OpenStack), **AI/ML** (LLM fine-tuning, computer vision, RAG) and **Cybersecurity** (securing AI agents and MCP servers, hardening of models and inference endpoints, payment and infrastructure security). Experience in secure SaaS deployment, industrial automation and network optimization (5G, fiber optics). Co-founder at AlgoMakers.ai and IT manager for two North American companies. Able to deliver end-to-end projects: from development to cloud infrastructure, including the integration and securing of AI agents.

EXPERIENCE

Summer Internship – Development – Softavera (Aures Konnect) ☞ Tunisia ■ Summer 2025
- Discovery of software production cycles, collaborative tools and coding best practices; participation in development and integration tasks.

Summer Internship – IT Team Leader – NeuroProdige (Quebec, Canada) ☞ Remote ■ 3 months, 2025
- Leadership of the IT team: task distribution, code review, and delivery follow-up.
- Development of the company website and of its cloud architecture (VPS/Nginx, TLS, CI/CD, backups).
- Securing of the payment flow: gateway integration, webhook signature verification, encryption of keys and sensitive data.
- Automation of business processes: CRM workflows, webhooks, notifications, and internal reporting.

Cloud, AI & Security Engineer – Freelance (Upwork) ■ 2025 – Present
- Development of AI voice agents and chatbots for various clients (OpenAI integration, RAG, MCPs).
- Fine-tuning of open-source LLMs (Qwen, Kimi) on domain-specific corpora; local or VPS deployment.
- Creation of a crypto market scanner with notification system and auto-trading via broker APIs.
- SaaS platform generating video ads from a product link (Streamlit, FFmpeg, secure Nginx deployment).
- Security audits of client deployments: server and API hardening, secret management, access control.

Co-founder & Marketing Manager – AlgoMakers.ai (Saudi Arabia) ■ 2025 – Present
- Management of a 9-person team: lead extraction bots, appointment scheduling, qualified chatbots.
- Deployment of cloud infrastructure (AWS, VPS) for quantitative trading services.

IT Manager – NeuroProdige & SoluClient (Quebec, Canada) ■ 2025 – Present
- Full administration of GHL environments: automations, webhooks, CRM integrations.
- Setup of secure servers (Nginx reverse proxy, encryption, protection of sensitive data).
- Operational continuity supervision for two fully remote companies.

CYBERSECURITY, CLOUD & AI PROJECTS

Multi-Layer Security Framework for MCP Servers
- Defense-in-depth architecture for MCP servers: OAuth2/JWT authentication, tokens scoped per tool, least-privilege authorization per client.
- Isolation layer: tools executed in Docker sandboxes, allow-list of commands and domains, quotas and rate limiting per session.
- Prompt-injection and exfiltration protection: I/O sanitization, strict schema validation of tool calls, secret redaction.
- Traceability: mTLS between agent and servers, signed audit logs, centralized secret management, anomaly alerting (Grafana).

Security of AI Models (LLM & Computer Vision)
- Hardening of inference endpoints: Nginx reverse proxy with TLS, API key encryption and rotation, rate limiting, anti-abuse filtering.
- Guardrails layer: jailbreak and malicious prompt detection, PII redaction before/after inference, multi-level validation of outputs.
- Protection of model assets: encrypted weights at rest, signed artifacts, mitigation of model extraction and data poisoning.
- Red-teaming: adversarial test suite (jailbreaks, adversarial examples on YOLOv8) and security scoring before each deployment.

Academic & Research Projects – ENETCOM (Sfax, Tunisia) ■ 2025 – 2026
- **SIVAR** : Real-time anomaly detection (YOLOv8, YOLOv8-Pose, Flask, Flutter, Firebase).
- **Smart 5G QoS Dashboard** : AI monitoring of a 5G network in Sfax (Streamlit, Isolation Forest, regression, Folium).
- **MTM-2 Analysis Generator** : Automated time and motion analysis (GPT fine-tuning, RAG, multi-level validation, Azure deployment).

PERSONAL PROJECTS & SAAS

- **Crypto auto-trading platform** : Endpoint security with Nginx, API key encryption, real-time database.
- **Affiliate marketing automation** : Extraction of winning campaigns and automated relaunch on other platforms.
- **Shopify e-commerce sites** : Custom development in Liquid code, SEO and conversion optimization.
- **Full SaaS deployments** : From architecture (Kubernetes, Docker) to secret management and monitoring (Prometheus, Grafana).

EDUCATION

Telecommunications Engineering Degree 2024 – 2026
ENET'COM – National School of Electronics and Telecommunications of Sfax (Tunisia)

Preparatory Cycle – Physics & Chemistry 2022 – 2024
IPEIM, Monastir (Tunisia)

SKILLS

- **Cybersecurity :** Server & API hardening, TLS/mTLS, OAuth2/JWT, payment & webhook security, secret management, sandboxing, penetration testing, LLM/agent security (prompt injection, jailbreak).
- **Cloud & DevOps :** Kubernetes, Docker, AWS, OpenStack, VPS (Ubuntu), Termius, Nginx (reverse proxy, TLS), GitHub Actions.
- **AI / Machine Learning :** LLM fine-tuning (Qwen, Kimi, GPT), RAG, MCPs, YOLOv8, PyTorch, Hugging Face, scikit-learn, Streamlit.
- **Networking & Telecom :** 5G/4G/GSM, fiber optics, Cisco Packet Tracer (CCNA1 & CCNA2), network security fundamentals (Google Cert).
- **Development & Algorithms :** Python, Flask, React, Flutter, Shopify Liquid, APIs (REST, WebSocket), databases (PostgreSQL, Firebase, SQLite).
- **Automation & Optimization :** Bots (lead extraction, messaging), scraping, GHL workflows, webhooks, ad campaign optimization.

CERTIFICATIONS

- CCNA 1 & CCNA 2 – Cisco
- HCIA-Datacom – Huawei (networking)
- eJPT – Junior Penetration Tester (INE / eLearnSecurity)
- Cybersecurity Fundamentals – Google
- SIVAR project certificate (real-time AI recognition)

LANGUAGES

Arabic : Native **French :** Fluent **English :** Fluent (technical)

LEADERSHIP & PROJECTS

Co-founder & Marketing Manager – AlgoMakers.ai (Saudi Arabia), 2024 – Present

IT Manager – two companies – NeuroProdige / SoluClient (Canada), 2024 – Present

Active member – ENETCOM Tech Clubs (Networking, AI, Cloud)